

Hands-on Lab:Working with a real world data-set using SQL in MySQL using phpMyAdmin



Estimated time needed: 20 minutes

In this lab, you will learn how to create tables and load data in the MySQL database service using the phpMyAdmin graphical user interface (GUI) tool.

Software Used in this Lab

In this lab, you will use [MySQL](#). MySQL is a Relational Database Management System (RDBMS) designed to efficiently store, manipulate, and retrieve data.



To complete this lab you will utilize MySQL relational database service available as part of IBM Skills Network Labs (SN Labs) Cloud IDE. SN Labs is a virtual lab environment used in this course.

Database Used in this Lab

`mysql_learners` database has been used in this lab.

Here you will be creating and inserting data into the below mentioned 2 tables

- 1.chicago_public_schools
- 2.chicago_socioeconomic_data

Here we will be using 2 dump files for this purpose.

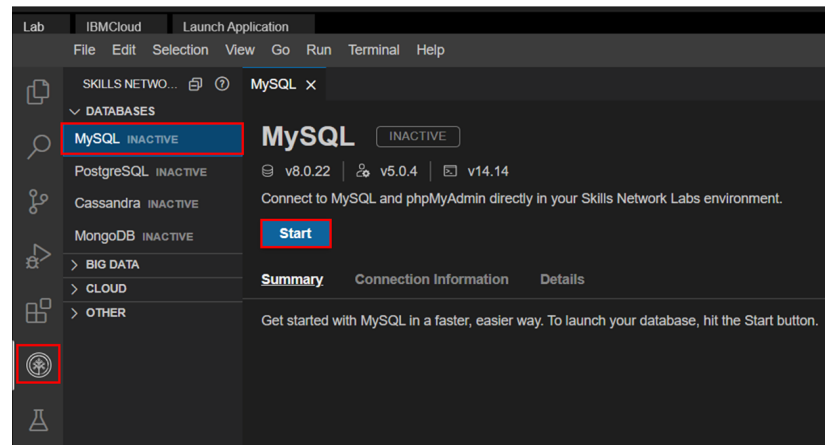
[chicago_public_schools](#)

[chicago_socioeconomic_data](#)

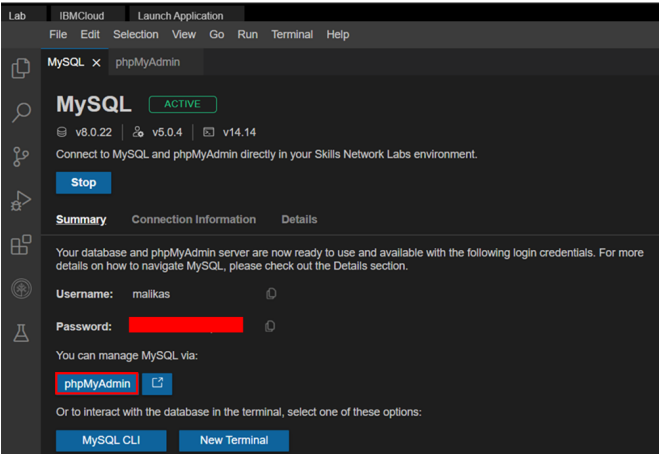
Task A: Create a database

1. Click on **Skills Network Toolbox**. In **Database** section, click **MySQL**.

To start the MySQL, click **Start**.



2. Once **MySQL** has started, click on **phpMyAdmin** button to open **phpMyAdmin** in the same window.



3. You will see the phpMyAdmin GUI tool.

phpMyAdmin

Recent Favorites

New

- information_schema
- mysql
- performance_schema
- sakila
- sys

Server: mysql:3306

Databases SQL Status User accounts Export Import Settings Binary log Replication Variables Charsets Engines

General settings

Server connection collation: utf8mb4_unicode_ci

[More settings](#)

Appearance settings

Language: English

Theme: pmahomme

Database server

- Server: mysql via TCP/IP
- Server type: MySQL
- Server connection: **SSL is not being used**
- Server version: 8.0.22 - MySQL Community Server - GPL
- Protocol version: 10
- User: root@172.18.0.2
- Server charset: UTF-8 Unicode (utf8mb4)

Web server

- Apache/2.4.38 (Debian)
- Database client version: libmysql - mysqlnd 7.4.15
- PHP extension: mysqli curl mbstring
- PHP version: 7.4.15

phpMyAdmin

- Version information: 5.0.4, latest stable version: 5.1.0
- [Documentation](#)
- [Official Homepage](#)
- [Contribute](#)
- [Get support](#)
- [List of changes](#)
- [License](#)

4. In the tree-view, click **New** to create a new empty database. Then enter **MySQL_Learners** as the name of the database and click **Create**.
The encoding will be left as **utf8mb4_0900_ai_ci**. UTF-8 is the most commonly used character encoding for content or data.
Proceed to Task B.

DatabasesSQLStatusUser accountsExportImportSettingsBinary logReplicationVariablesCharsetsMore

Databases

Create database ⓘ

Mysql_learners

utf8mb4_0900_ai_ci

Create

Database	Collation	Master replication	Action
☐ information_schema	utf8_general_ci	✔ Replicated	Check privileges
☐ mysql	utf8mb4_0900_ai_ci	✔ Replicated	Check privileges
☐ performance_schema	utf8mb4_0900_ai_ci	✔ Replicated	Check privileges
☐ sys	utf8mb4_0900_ai_ci	✔ Replicated	Check privileges
Total: 4			

⬆

☐ Check all

With selected:

[Drop](#)

⚠ Note: Enabling the database statistics here might cause heavy traffic between the web server and the MySQL server.

• Enable statistics

Load the dump files one by one into the database **Mysql_learners** by clicking the **Import** tab and choose the file.
Click on **Go** button.

Importing into the database "Mysql_learners"

File to import:

File may be compressed (gzip, bzip2, zip) or uncompressed.
A compressed file's name must end in **.[format].[compression]**. Example: **.sql.zip**

Browse your computer:

Choose File

chicago_pu...schools.sql (Max: 2,048KiB)

You may also drag and drop a file on any page.

Character set of the file:

utf-8

Partial import:

☒ Allow the interruption of an import in case the script detects it is close to the PHP timeout limit. *(This might be a good way to import large files, however it can break transactions.)*

Skip this number of queries (for SQL) starting from the first one:

0

Other options:

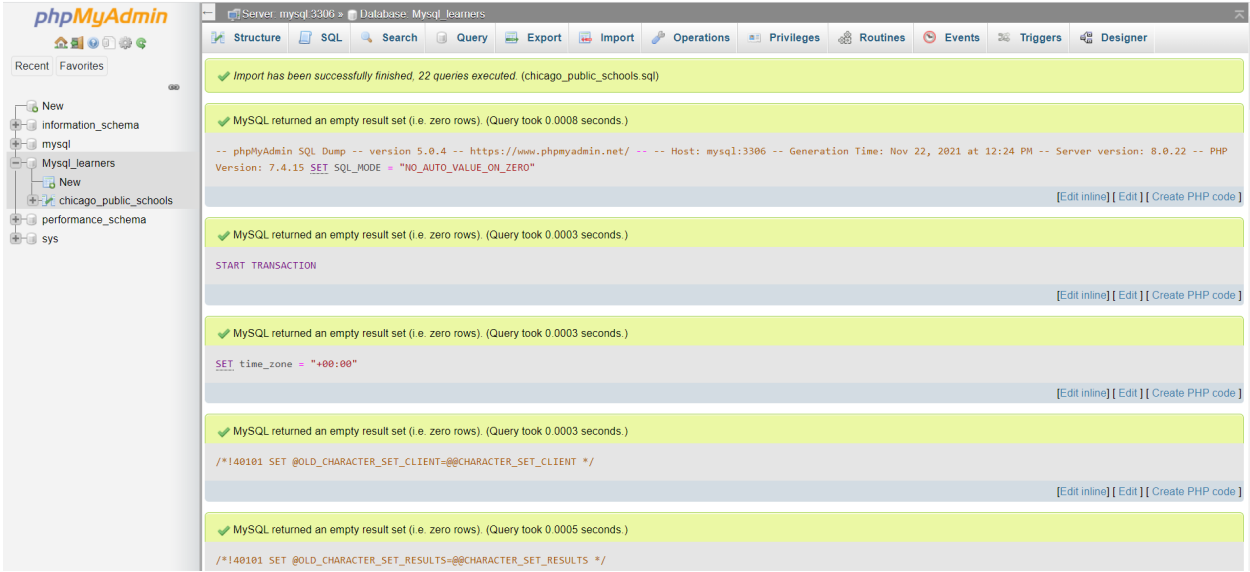
☒ Enable foreign key checks

Format:

Console

4 of 8

1/11/2026, 7:23 PM



The tables are created and the data is loaded successfully. Repeat the same operation with the other dump file to create and load the table.

Problems

Problem 1

How many Elementary Schools are in the dataset?

▼ Click here for a hint
Which column specifies the school type e.g. 'ES', 'MS', 'HS'? ('Elementary School, Middle School, High School')

▼ Click here for another hint
Does the column name have mixed case, spaces or other special characters?
If so, ensure you use ` ` symbol around the 'Name of the Column'

▼ Click here for the solution.
Solution:
select count(*) from chicago_public_schools where `Elementary, Middle, or High School` ='ES'
Correct answer: 462

Problem 2

What is the highest Safety Score?

▼ Click here for a hint
Use the MAX() function

▼ Click here for the solution.

```
select MAX(Safety_Score) AS MAX_SAFETY_SCORE from chicago_public_schools;
Correct answer: 99
```

Problem 3

Which schools have highest Safety Score?

▼ Click here for the solution.

Solution:
In the previous problem we found out that the highest Safety Score is 99, so we can use that as an input in the where clause:
select Name_of_School, Safety_Score from chicago_public_schools where Safety_Score = 99
or, a better way:
select Name_of_School, Safety_Score from chicago_public_schools where
Safety_Score= (select MAX(Safety_Score) from chicago_public_schools)
Correct answer: several schools with with Safety Score of 99.

Problem 4

What are the top 10 schools with the highest "Average Student Attendance"?

▼ Click here for the solution.

```
select Name_of_School, Average_Student_Attendance from chicago_public_schools order by Average_Student_Attendance desc limit 10;
```

Problem 5

Retrieve the list of 5 Schools with the lowest Average Student Attendance sorted in ascending order based on attendance

▼ Click here for the solution.

```
SELECT Name_of_School, Average_Student_Attendance
from chicago_public_schools
order by Average_Student_Attendance
LIMIT 5;
```

Problem 6

Now remove the "%" sign from the above result set for Average Student Attendance column

▼ Click here for a hint.

Use the REPLACE() function to replace 'X' with ''

▼ Click here for the solution.

```
SELECT Name_of_School, REPLACE(Average_Student_Attendance, 'X', '')
from chicago_public_schools
order by Average_Student_Attendance
LIMIT 5
```

Problem 7

Which Schools have Average Student Attendance lower than 70%?

▼ Click here for a hint.

The datatype of the "Average_Student_Attendance" column is varchar.
So you cannot use it as is in the where clause for a numeric comparison.
First use the CAST() function to cast it as a DECIMAL or DOUBLE
e.g. CAST("Column_Name" as DOUBLE)

► Click here for another hint

Don't forget the 'X' age sign needs to be removed before casting

► Click here for the solution.

Solution:

```
SELECT Name_of_School, Average_Student_Attendance from chicago_public_schools
where CAST(REPLACE(Average_Student_Attendance, 'X', '' ) AS DOUBLE) < 70
order by Average_Student_Attendance
```

Problem 8

Get the total College Enrollment for each Community Area

► Click here for a hint.

Verify the exact name of the Enrollment column in the database
Use the SUM() Function to add up the Enrollments for each Community Area

► Click here for another hint.

Don't forget to group by the Community Area

► Click here for the solution.

Solution:

```
select Community_Area_Name, sum(College_Enrollment) AS TOTAL_ENROLLMENT
from chicago_public_schools
group by Community_Area_Name
```

Problem 9

Get the 5 Community Areas with the least total College Enrollment sorted in ascending order

► Click here for a hint.

Order the previous query and limit the number of rows you fetch

► Click here for the solution.

Solution:

```
select Community_Area_Name, sum(College_Enrollment) AS TOTAL_ENROLLMENT
from chicago_public_schools
group by Community_Area_Name
order by TOTAL_ENROLLMENT asc
LIMIT 5
```

Problem 10

List 5 schools with lowest safety score.

► Click here for the solution.

```
Solution:
SELECT name_of_school, safety_score
FROM chicago_public_schools
ORDER BY safety_score
LIMIT 5
```

Problem 11

Get the hardship index for the community area which has College Enrollment of 4368

► Click here for the solution.

```
Solution:
NOTE: For this solution to work the CHICAGO_SOCIOECONOMIC_DATA table
      as created in the last lab of week 3 should already exist
select hardship_index from chicago_socioeconomic_data CD,
chicago_public_schools CPS where CD.community_area_number = CPS.community_area_number and college_enrollment = 4368
```

Problem 12

Get the hardship index for the community area which has the school with the highest enrollment.

► Click here for the solution.

```
Solution:
select community_area_number, community_area_name, hardship_index from chicago_socioeconomic_data
where community_area_number in
( select community_area_number from chicago_public_schools where COLLEGE_ENROLLMENT = (select MAX(COLLEGE_ENROLLMENT) from chicago_public_schools) );
```

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